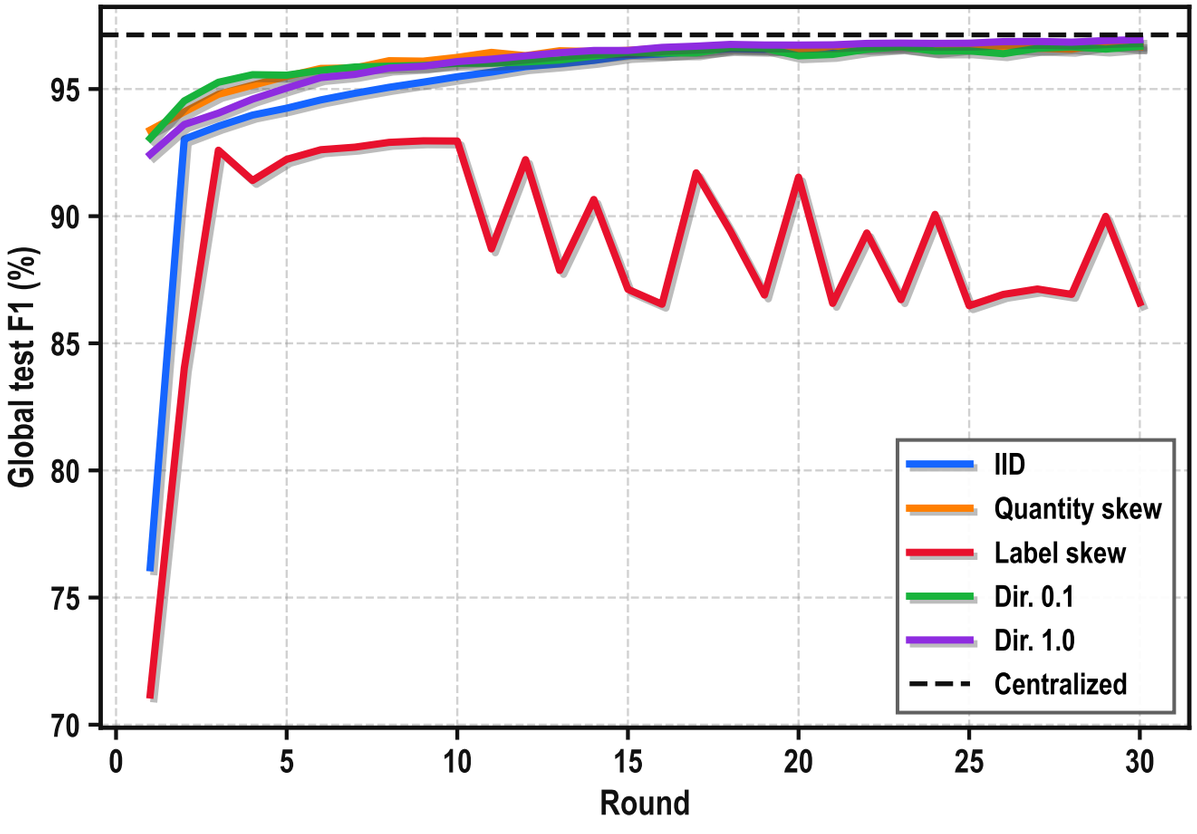
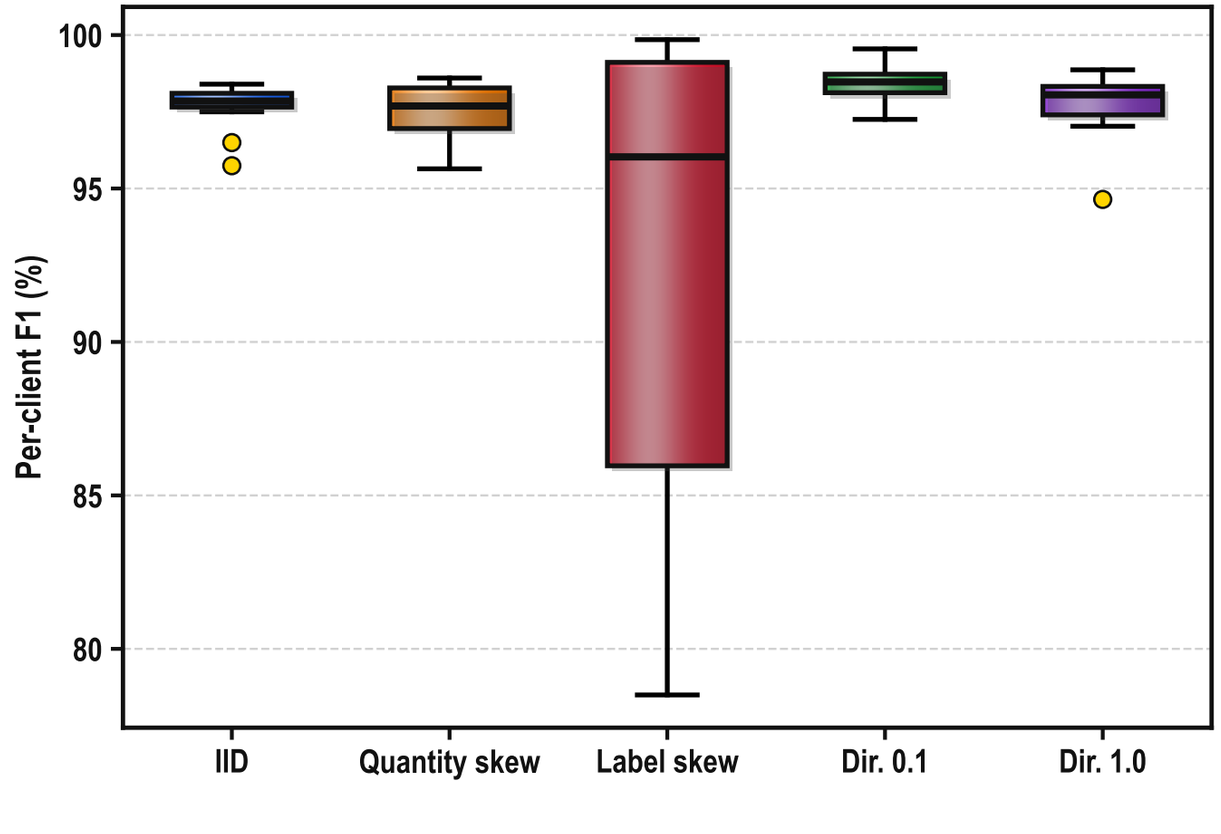
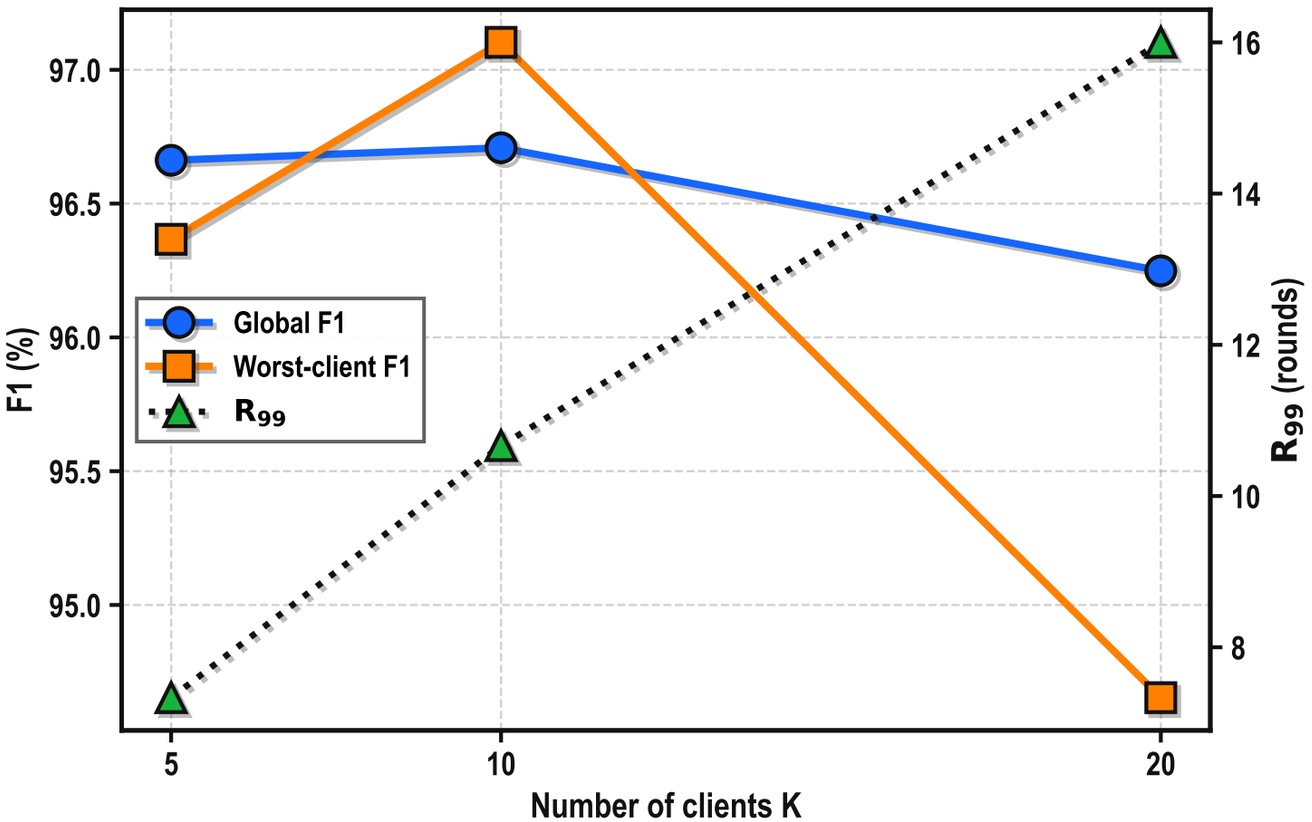
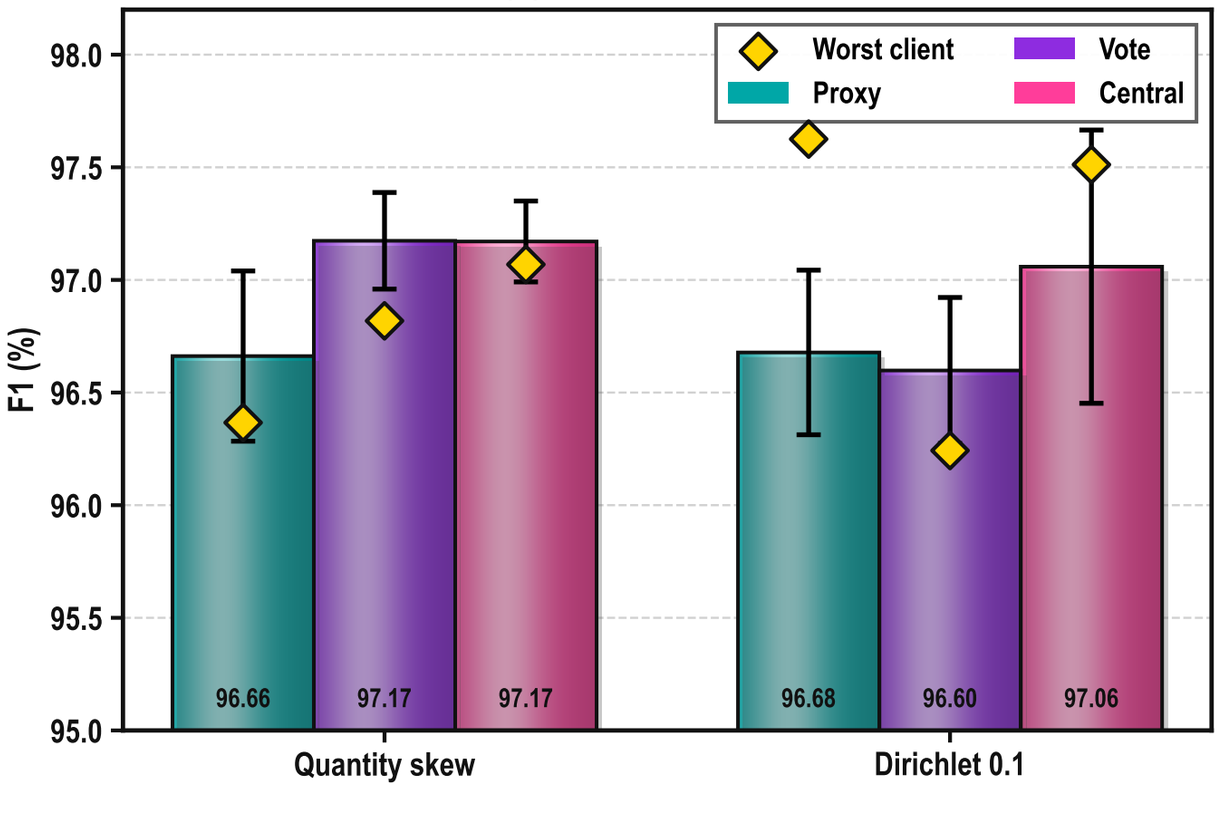

(a) Convergence (FedAvg, K=5)
Global test F1 (%)
Round
IID
Quantity skew
Label skew
Dir. 0.1
Dir. 1.0
Centralized


(b) Per-client F1 (all clients x 3 seeds)
Per-client F1 (%)
IID
Quantity skew
Label skew
Dir. 0.1
Dir. 1.0


(c) Client count (quantity skew)
F1 (%)
R99 (rounds)
Global F1
Worst-client F1
R99
Number of clients K


(d) Mask construction
F1 (%)
Worst client
Proxy
Vote
Central
96.66
97.17
97.17
96.68
96.60
97.06
Quantity skew
Dirichlet 0.1